Investigations of Performance of Fire Fighting Robot in Mining Industry

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Abstract— Mining operations are carried out in an environment that is susceptible to fire risks and represents a substantial risk to human life and property. Traditional firefighting approaches put human responders in dangerous situations and create urgent demands for safer, automated alternatives. This study presents the construction, construction and performance analysis of autonomous firefighting robots driven by Arduino and developed specifically for mining applications. The robot has three flame sensor used for accurate fire detection, water pumps to put fire out, motor drivers for mobility, and servo motors for water distribution. The robot has key performance indicators such as fire detection accuracy, autonomous navigation, fire extinguishing under hard mining conditions, and operational reliability were checked. Also experiments were performed to replicate mining scenarios and checking the robot's capabilities for autonomous navigation and fire extinguishing. The results show that developed robotic systems can make a significant contribution to improving fire security in mines, providing a reliable way to reduce risk, reduce manpower, save human lives and protect valuable assets. This paper also describes potential improvements and future developments aimed at increasing the power and autonomy of robots.

Keywords-- Firefighting Robot, Mining Safety, Autonomous Navigation, Arduino-Based System,

I. INTRODUCTION

The mining sector, is one of the most important pillar of global economic progress, works in natural and dangerous environments, particularly at high risk of fire. Underground mining operations are particularly vulnerable due to small spaces, presence of flammable materials, and concentration of explosive gases. These factors create a volatile environment in which traditional method of fire extinguishing exposes extreme dangers such as toxic gases, collapse, poor visibility and limited access. As a result, there is an urgent need for advanced technology to improve fire prevention and response strategies and ensure the protection of human life and critical mining infrastructure.

This project designs a fire robot that can detect and put out fires without human intervention that uses Arduino based technology. By integrating a network of sensors, engines and intelligent control systems, the robot can identify fire sources and immediately adhere to them and suppress them. This ability places fire robots as a key asset in minimizing fire-related risks and improving general security measures. The proposed system is designed to vanish when small flames are autonomously captured and sailed towards the fire service. A few flame sensors are used to cover different directions. Left sensor monitors the left side, the centre sensor monitors the front and right sensors monitor the right side. When a fire is detected, the robot activates the mechanism of extinguishing fire, moving towards the fire and suppressing the flames. This method ensures that a fire will be dealt with immediately, reducing further damage.

The most important features of the fire fighting robot include real-time fire monitoring, early detection, and automated

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navigation aimed at preventing accidents and protecting human life. We evaluate factors such as operational efficiency, effectiveness in fire control, thermal durability, and the ability to autonomously navigate complex underground environments. By examining these aspects under simulated actual mining conditions, this research aims to highlight the limitations of current firefighting technology and propose improvements to the different challenges of mining.

II. OBJECTIVES

The main aim of this research is to put the fire out using an Arduino based fire fighting robot in mining industry.

The most specific objectives of this research are as follows:

1. **Design and Development**
   - To design a robot using Arduino that can detect and put out small fires found in mines.
   - To place three flame sensors on the robot so it can detect fire from different directions correctly.
2. **Fire Detection and Suppression System**
   - The robot should be able to detect fire using flame sensors and determine its location.
   - It should include a fire suppression system such as a water pump, CO2 extinguisher, or a fan to put out the fire.
3. **Autonomous Movement and Control**
   - To create smart movement systems that help the robot find fires on its own while avoiding obstacles in the tunnels.
   - To test the robot's ability to work without help in changing and unpredictable mining environments.
4. **Safety and Reliability**
   - To check if the robot's parts can handle high temperatures and rough conditions underground.
   - To make sure the robot works steadily and reliably over long periods inside the mine.
5. **Suggestions for Improvement**
   - To find and understand any problems faced during the robot's design, building, and testing.
   - To give ideas for making the robot better, stronger, and more useful for firefighting in mines.
6. **Fire Safety measures**
   - The robot should have safety measures like emergency shutdown and heat resistive materials.

III. RESEARCH METHODOLOGY

The project deeply focuses on building a robot that can find and put out fires in small areas without needing help from people. The work was done in following three main steps:

1. **Robot Design:**

The first and most essential step was designing the robot's body. Since the robot needs to operate in narrow and rugged mining environments, the design had to be both compact and sturdy. The aim was to ensure smooth movement even in confined and challenging areas.

2. **Hardware Setup:**

Next, we focused on selecting suitable hardware components and assembling them correctly. Key components included BO motors for mobility, flame sensors to detect fire, and a small water pump system for firefighting. Each part was carefully integrated to work seamlessly within the robot.

3. **Programming and Control:**

The final step involved coding and uploading the program to the Arduino UNO. The software enables the robot to react promptly when fire is detected — by activating the water pump to spray water and suppress the flames automatically.

After all the processes were completed, tests were done to see how well the robot could find and put out fires. These tests helped checking if the robot was working as planned.

A. Design Structure:

This parts gives the designing of robot and how the robot is made and how it works. There are various components used while making the robot such as flame sensors (IR), servo motors, a small water pump, a motor driver, a mini breadboard, BO motors, rubber wheels, a processor, and a communication module to connect with the Arduino.

The robot is designed to do four main tasks:

1. **Start-up** – When the robot is turned on, all the flame sensors start working together.
2. **Monitoring** – The robot keeps checking its surroundings to find signs of fire, like high temperatures or flames.
3. **Movement** – If a fire is found, the navigates towards it using BO motors.
4. **Firefighting** – Once it is reached at the location of fire, the robot uses the servo motors and water pump to spray water and put the fire out.

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1. ***Circuit Diagram:***

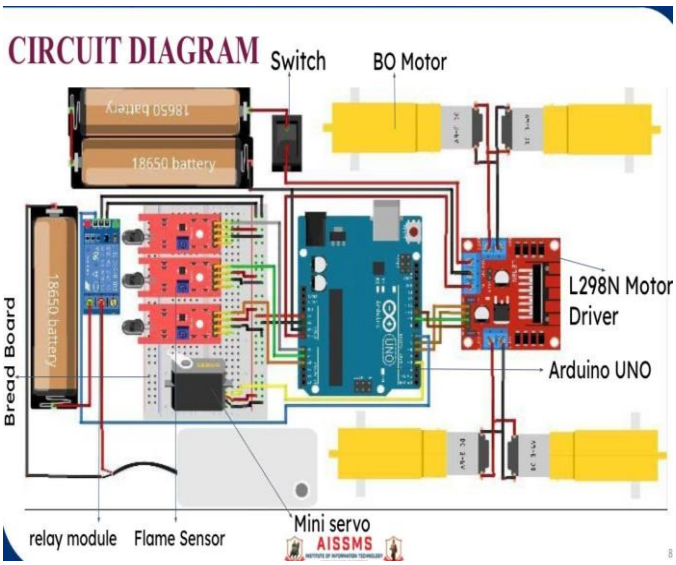

Figure 1

B. Hardware Used:

1) Arduino Uno

Figure 2

The Arduino Uno plays a central role in robotics embedded Projects. It processes input signals from the flame sensors, executes the program, and sends commands to the motors, servo, and water pump.

2) Flame Sensor:

Figure 3

The flame sensor is used to spot fire by detecting the infrared light that flames give off. It keeps checking the area around the robot and sends a signal to the Arduino when it sees a fire. Using this signal, the robot can figure out where the fire is and move toward it.

3) Breadboard:

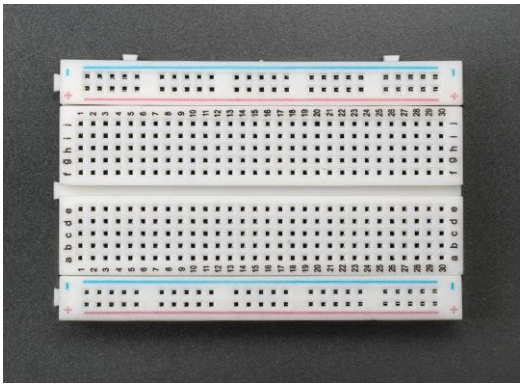

Figure 4

The breadboard is a tool that lets you connect the robot's electronic parts without soldering. It makes it easy to try out different setups, fix problems, and make quick changes, which is great when building and testing the robot.

4) Relay Module:

Figure 5

The relay module allows the Arduino to control high-power components, such as the water pump, by acting as an intermediate switch. Since the Arduino itself cannot supply enough current to run the water pump, the relay ensures that power is only delivered when necessary.

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5) Motor Driver:

Figure 6

The motor driver is responsible for controlling the robot's movement by regulating the BO motors. It allows the Arduino to adjust the direction and speed of the motors, enabling the robot to move towards the fire source.

6) Water Pump:

Figure 7

When the flame sensor detects a fire, the Arduino activates the water pump via the relay module. The pump then sprays water at the fire, effectively suppressing the flames.

7) Mini Servo:

Figure 8

The mini servo motor provides precise control over small movements, such as adjusting the direction of the water nozzle or reorienting flame sensors. This fine-tuned control helps the robot accurately target the fire source and improve the efficiency of the firefighting process.

C. Programming:

1. Upload the required code to the Arduino board to manage the flame sensors, motor driver, and water pump motor.
2. In the code, continuously monitor the outputs from the flame sensors to identify the presence of fire.
3. Based on the sensor data, control the motor driver to allow the robot to move autonomously in the desired direction.
4. Upon detecting fire, activate the water pump motor by triggering the Arduino's digital pin connected to the transistor's base, initiating the water flow.

new.ino

```
#include <Servo.h>

// Define flame sensor pins
#define FLAME_SENSOR_LEFT A0
#define FLAME_SENSOR_CENTER A1
#define FLAME_SENSOR_RIGHT A2

// Define motor driver pins
#define MOTOR_L1 5  // Left Motor Forward
#define MOTOR_L2 6  // Left Motor Backward
#define MOTOR_R1 9  // Right Motor Forward
#define MOTOR_R2 10 // Right Motor Backward

// Define relay for water pump
#define RELAY_PIN 7

// Define servo motor pin
#define SERVO_PIN 3

// Create Servo object
```

Figure 9
Arduino IDE Program

IV. RESULTS AND DISCUSSION

The fire sensor outputs a HIGH signal when detecting fire and a LOW signal when there is no fire. The system continuously checks the sensor readings to detect the presence of fire. If no fire is detected, the motors remain stationary by setting all control pins to HIGH. When fire is detected, the robot will move towards the fire by activating the corresponding motor. Once the robot is positioned directly in front of the fire, the left and right sensors will no longer detect the flame as the robot is facing the source. A variable named "fire" will then be used to trigger the function responsible for extinguishing the fire. Once the "fire" variable is set to TRUE, the fire-fighting function will execute the "put_off_fire" routine, which will continue until the fire is completely extinguished.

In the "put_off_fire()" function, the robot stops by turning on all the motor pins. Then, the water pump starts to spray water from the tank. At the same time, the servo motor moves the water container to spread the water evenly over the fire.

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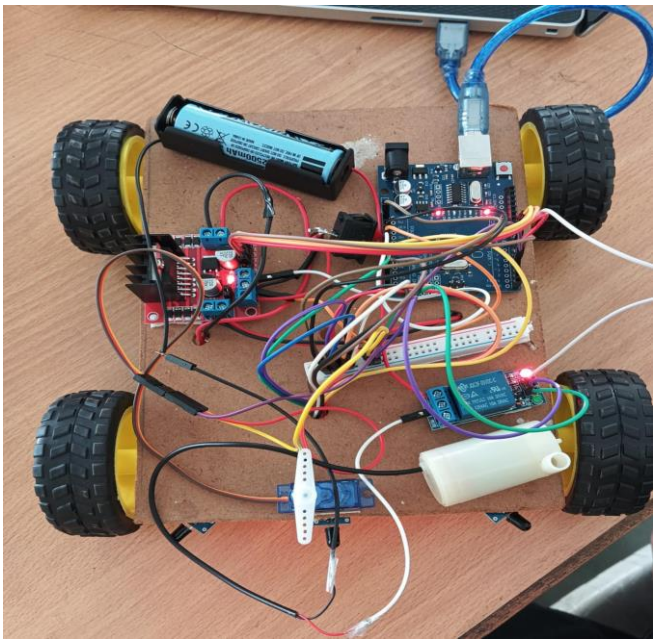

Figure 10
Working Model

V. FUTURE SCOPE

1. **Improved Fire Labeling Algorithm:**

   The development of advanced algorithms for fire detection, including integration of machine learning models trained on various brand types, can greatly improve system accuracy and minimize false alarms. These algorithms can use different sensor data for more reliable detection.

2. **Multi-sensor integration:**

   Expansion of the sensor suite to additional detectors such as road surfaces (for toxic steam), thermal sensors (for temperature profiling), and visual cameras improves the robot's ability to assess situations. This multimodal approach gives a more comprehensive understanding of the environment and improves the overall effectiveness of the robot.

3. **Improved Autonomous Navigation:**

   Including advanced navigation and localization algorithms such as simultaneous localization and assignment (SLAM) enables robots to navigate complex environments more efficiently. These technologies allow real-time mapping of the site and help robots avoid obstacles while firefighting tasks are being performed.

4. **Dynamic Fire Extinction System:**

   Adaptive removal system design The corresponding fire suppression method can be selected based on the type of fire, and strength is important. These systems can use a variety of active ingredients, such as water, bubbles, dry chemicals, and even specialized fire prevention agents, so that the robot can effectively tackle a variety of fire scenarios.

VI. CONCLUSION

In summary, the inclusion of fire fighting robots in mining industry signifies a major breakthrough in improving safety and ensuring uninterrupted productivity. These systems are engineered to handle hazardous situations that are typically too risky for workers, providing a safer alternative for managing fire emergencies deep underground.

Equipped with flame sensors, BO motors, and fire suppression tools, robotic units can swiftly evaluate and address fire risks in areas with limited visibility, poor air quality, and restricted access. As these technologies continue to advance, future models are anticipated to be even more autonomous, intelligent, and adept at handling a wide array of intricate situations.

The ability for multiple robots to work together during fire fighting operations enhances efficiency, allowing for large-scale strategies that enhance coverage and decrease response times. Moreover advancements in communication technology, battery efficiency, and environmental consciousness will contribute to the enhanced reliability and functionality of these robotic systems.

As expenses decrease and the technology becomes more readily available, the mining industry is expected to witness a wider implementation of robotic fire fighting solutions. These advancements facilitate a transition towards proactive safety measures, minimizing downtime, safeguarding assets, and most importantly, preserving lives. By incorporating robotics into emergency preparedness plans, mining operations can enhance their resilience, efficiency, and ability to handle fire-related risks.

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